

① What is the mass of the atmosphere?

ans) Atmosphere := 80-10000 km^h

Pressure = $\frac{\text{Force}}{\text{Area}}$ } force exerted by
an overhead column

$$1 \text{ Pa} = 1 \text{ Nm}^{-2} \quad \text{of atmosphere per}$$

$$1 \text{ J} = 1 \text{ Nm} = 1 \text{ Pa m}^3 \quad \text{unit area.}$$

Atmospheric pressure at mean sea level

$$\left[\begin{aligned} &= 1 \text{ atm} \\ &= 1.0125 \times 10^5 \text{ Pa} \end{aligned} \right]$$

$$\text{Force} = M_a g$$

$$\text{Pressure} = \frac{M_a g}{\text{Area of Earth}} = \frac{M_a g}{4\pi R_e^2} \quad (\text{assuming earth is spherical})$$

$$1.0125 \times 10^5 = \frac{M_a (9.8)}{4\pi (6.4 \times 10^6)^2}$$

$$M_a \approx \frac{10^5 \times 4 \times 3.14 \times (6.4 \times 10^6)^2}{10}$$

$$\approx \underline{\underline{5.148 \times 10^{18} \text{ kg}}}$$

② What is the molecular weight of air?

ans) A molecule of air — undefined
↓
combination of 1 or more elements
in a fixed proportion.

This question is meaningful when the
composition of air is well constrained.

— a hypothetical molecule of air

Ideal gas Equation :

$$pV = nRT \quad \left. \begin{array}{l} \text{ } \end{array} \right\} \text{applicable because}$$

of ambient pressure & temperature conditions.

Composition of air:

	%	Mixing ratio
N ₂	78	0.78
O ₂	21	0.21
Ar	0.9	0.09
Others	0.1	0.01

Molecular weight of air = weighted avg.
 = $\sum x_m x$

$$\approx 0.78(28) + 0.21(32) + 0.01(40)$$

$$\approx \underline{\underline{28.96 \text{ g mol}^{-1}}}$$

Q. What is the molecular weight of air?
 A. A molecule of air is a mixture of various elements in a fixed proportion. This question is meaningless when the composition of air is not constant. The composition of air is not constant as it varies from place to place. The composition of air is not constant as it varies from place to place. The composition of air is not constant as it varies from place to place.

30/8

EES 202

T2.

- ① A weather station reports $T = 293\text{K}$, $RH = 50\%$ at sunset. Assuming that P_{H_2O} remains constant, by how much must the temperature drop over the course of the night in order for the fog to form?

ans) $RH\% = 50\% = \frac{P_{H_2O}}{P_{H_2O, SAT}} \times 100 \Rightarrow P_{H_2O} = \frac{P_{H_2O, SAT}}{2}$

$P_{H_2O, SAT} = A \exp \left[B \left(\frac{1}{T} - \frac{1}{T_0} \right) \right]$ $A = 6.11 \text{ hPa}$
 $B = -5310 \text{ K}$

$= (6.11 \times 10^3 \text{ Pa}) \exp \left[-5310 \left(\frac{1}{293} - \frac{1}{273} \right) \right]$ $T_0 = 273\text{K}$

$= 2.3 \times 10^3 \text{ Pa}$

$P_{H_2O} = \frac{P_{H_2O, SAT}}{2} = 1.15 \times 10^3 \text{ Pa}$

At dew point $T' P_{H_2O} = P'_{H_2O, SAT}$

$P'_{H_2O, SAT} = A \exp \left[B \left(\frac{1}{T'} - \frac{1}{T_0} \right) \right]$

$\frac{1}{T'} = \frac{1}{B} \ln \left(\frac{P'_{H_2O, SAT}}{A} \right) + \frac{1}{T_0}$

$= 3.54 \times 10^{-3} \text{ K}^{-1}$

$T' = 282.17 \text{ K}$

$\Delta T = T - T' = 293 - 282.17$

$= 10.83 \text{ K}$

The temperature should drop over by 10.83K over the course of night in order for the fog to form.

② What is the mass concentration of water vapor (g. H_2O per m^3 of air) in a liquid-water cloud at a temperature of 273K? Considering that the liquid water mass concentration in a cloud ranges typically from 0.1 to 1 g liquid water per m^3 of air, is most of the water in a cloud present as vapor or as liquid?

ans) $p_{H_2O, SAT} = A \exp \left[B \left(\frac{1}{T} - \frac{1}{T_0} \right) \right]$ $A = 6.11 \text{ hPa}$
 $B = -5310 \text{ K}$
 $T_0 = 273 \text{ K} = T$
 $= 6.11 \text{ hPa}$

From Ideal gas eqn,

$$pV = nRT$$

$$\frac{n}{V} = \frac{p}{RT}$$

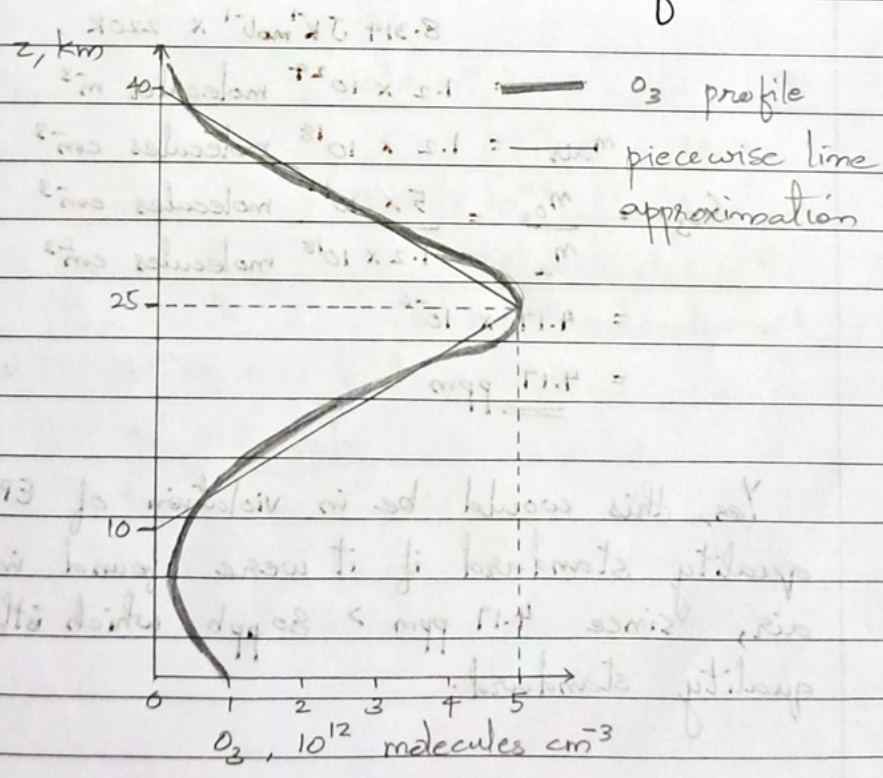
$$\frac{n M_{H_2O}}{V} = \frac{p M_{H_2O}}{RT}$$

$$= \frac{6.11 \times 10^2 \text{ Pa} \times 18 \text{ g mol}^{-1}}{8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 273 \text{ K}}$$

$$= 4.85 \text{ g m}^{-3}$$

Considering that the liquid water mass concentration in a cloud ranges typically from 0.1 to 1 g m^{-3} , most of the water in a cloud is present as vapor.

③ Consider the following typical vertical profile of ozone (O_3) number densities measured over the United States. Ozone is produced in the stratosphere (10-50 km altitude) by photolysis of O_2 and subsequent combination of atoms with O_2 . The stratospheric ozone layer protects life on earth by absorbing solar UV radiation and preventing this radiation from reaching the Earth's surface. Fortunately the O_3 layer is not in contact with the Earth's surface; inhalation of O_3 is toxic to humans and plants, and the U.S. Environmental Protection Agency (EPA) presently has an air quality standard of 80 ppbv O_3 not to be exceeded in surface air.



- (i) Calculate the mixing ratio of O_3 , at the peak of the O_3 layer ($z = 25$ km, $P = 35$ hPa, $T = 220$ K). Would this mixing ratio be in violation of EPA air quality standard if it were found in surface air? $z \rightarrow n_{O_3} = C_{O_3} n_a$

ans) At $z = 25$ km, $n_{O_3} = 5 \times 10^{12}$ molecules cm^{-3}
 $P = 35$ hPa, $T = 220$ K.

To calculate $C_{O_3} = ?$

$$n_{O_3} = C_{O_3} n_a, \quad n_a = \text{number density of air}$$

From Ideal gas eqn, $pV = nRT$

$$\frac{n_a V}{V} = \frac{p_a V}{RT}$$

$$n_{air} = \frac{p_a V}{RT}$$

$$= \frac{35 \times 10^2 \text{ Pa} \times 6.022 \times 10^{23}}{8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 220 \text{ K}}$$

$$= 1.2 \times 10^{24} \text{ molecules } m^{-3}$$

$$n_{air} = 1.2 \times 10^{18} \text{ molecules } cm^{-3}$$

$$C_{O_3} = \frac{n_{O_3}}{n_a} = \frac{5 \times 10^{12} \text{ molecules } cm^{-3}}{1.2 \times 10^{18} \text{ molecules } cm^{-3}}$$

$$= 4.17 \times 10^{-6}$$

$$= \underline{\underline{4.17 \text{ ppm}}}$$

Yes, this would be in violation of EPA air quality standard if it were found in surface air, since $4.17 \text{ ppm} > 80 \text{ ppb}$ which is the air quality standard.

Q Calculate the mixing ratio of O_3 in surface air ($z=0$ km, $p=1000$ hPa, $T=300$ K). Is it in compliance with the EPA air quality standard? Notice that the relative decrease in mixing ratio between 25 km and the surface is considerably larger than the relative decrease in number density. Why this? (2) $\rightarrow$ (10)

ans) At $z=0$ km, $n_{O_3} = 10^{12}$ molecules cm^{-3}

$$p = 1000 \text{ hPa}, T = 300 \text{ K}$$

To calculate $C_{O_3} = ?$ $n_{O_3} = C_{O_3} n_a$

$n_a =$ number density of air

From Ideal gas eqn, $pV = nRT$

$$\frac{n}{V} = \frac{p}{RT} \quad \text{or} \quad \frac{n_a V}{V} = \frac{p_a V}{RT}$$

$$n_a = \frac{p_a V}{RT}$$

$$= \frac{10^3 \times 10^2 \text{ Pa} \times 6.022 \times 10^{23}}$$

$$8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 300 \text{ K}$$

$$= 2.41 \times 10^{25} \text{ molecules m}^{-3}$$

$$C_{O_3} = \frac{n_{O_3}}{n_a} = \frac{10^{12} \times 10^6 \text{ molecules m}^{-3}}{2.41 \times 10^{25} \text{ molecules m}^{-3}}$$

$$= 4.15 \times 10^{-8}$$

$$= 41.5 \text{ ppb}$$

- (iii) The total number of O_3 molecules per unit area of Earth's surface is called the O_3 column and determines the efficiency with which the O_3 layer prevents solar UV radiation from reaching the Earth's surface. Estimate the O_3 column in the above profile by approximating the profile with the piecewise linear function shown as the thin solid line.
- ans) To calculate the no. of O_3 molecules cm^{-2}

$$\begin{aligned}
 \text{Area} &= \frac{1}{2} \times b \times h \\
 &= \frac{1}{2} \times 30 \text{ km} \times 5 \times 10^{12} \text{ molecules } cm^{-3} \\
 &= \frac{1}{2} \times 30 \times 10^5 \text{ km} \times 5 \times 10^{12} \text{ molecules } cm^{-3} \\
 &= 75 \times 10^{17} \text{ molecules } cm^{-2} \\
 &= \underline{7.5 \times 10^{18}} \text{ molecules } cm^{-2}
 \end{aligned}$$

- (iv) To illustrate how thin the ^{stratospheric} ~~atmosphere~~ O_3 layer actually is, imagine that all of the O_3 in the atmospheric column was brought to sea level as a layer of pure O_3 gas under standard conditions of temperature and pressure ($1.013 \times 10^5 \text{ Pa}$, 273 K). Calculate the thickness of the layer. $V = Ah$

ans) $p = 1.013 \times 10^5 \text{ Pa}$ $T = 273 \text{ K}$ $V = Ah$

$$\begin{aligned}
 pV &= pAh = nRT \quad \quad \quad n/A = 7.5 \times 10^{18} / A \text{ molecules } cm^{-2} \\
 h &= \frac{nRT}{pA} = \frac{7.5 \times 10^{18} \times 8.314 \times 273}{6.022 \times 10^{23} \times 1.013 \times 10^5 \times 10^{-4}} \\
 &= 2.79 \times 10^{-3} \text{ m} \\
 &= 2.79 \text{ mm}
 \end{aligned}$$

06/09 EES202

T3.

- ① Atmospheric CO_2 concentrations increases from 280 ppm to 365 ppm. What is the corresponding increase of carbon?

ans) $C_{\text{CO}_2} = n_{\text{CO}_2} / n_a = m_{\text{CO}_2} (M_a) / m_a (M_{\text{CO}_2})$

$\Delta C_{\text{CO}_2} \quad [n = m/M]$

$$\Delta C_{\text{CO}_2} = \frac{\Delta m_{\text{CO}_2} M_a}{m_a M_{\text{CO}_2}}$$

$$\Delta m_{\text{CO}_2} = \frac{\Delta C_{\text{CO}_2} m_a M_{\text{CO}_2}}{M_a}$$

$$= \frac{(365 - 280) \times 10^{-6} \times 5.2 \times 10^{18} \text{ kg} \times 12 \text{ g mol}^{-1}}{28.96 \text{ g mol}^{-1}}$$

$$= \frac{85 \times 5.2 \times 12 \times 10^{12}}{28.96} \text{ kg}$$

$$= 1.83 \times 10^{14} \text{ kg}$$

- ② Mercury is toxic and expensive. Replacing Hg with water in the barometer instrument, what is the length of the tube required? (Density of Hg = 13.6 g cm^{-3})

ans) $p = \rho g h \Rightarrow \rho_{\text{Hg}} g h_{\text{Hg}} = \rho_{\text{H}_2\text{O}} g h_{\text{H}_2\text{O}}$

$$h_{\text{H}_2\text{O}} = \frac{\rho_{\text{Hg}} h_{\text{Hg}}}{\rho_{\text{H}_2\text{O}}}$$

$$= \frac{13.6 \text{ g cm}^{-3} \times 76 \text{ cm}}{1 \text{ g cm}^{-3}}$$

$$= 10.34 \text{ m}$$

- ③ On the Mars, atmosphere is mainly CO_2 , the temperature is 220K and acceleration due to gravity is 3.7 m s^{-2} . What is the scale height?

ans) $H = \frac{RT}{Mg} = \frac{8.314 \text{ J mol}^{-1} \text{ K}^{-1} \times 220 \text{ K}}{10^{-3} \times 94 \text{ kg mol}^{-1} \times 3.7 \text{ m s}^{-2}}$
 $= 1.124 \times 10^4 \text{ m} = \underline{\underline{11.24 \text{ km}}}$

- ④ Calculate the mass density of air at the surface of the Earth using ideal gas law and assuming global average surface pressure (984 hPa) & temperature 288K.

ans) $pV = nRT = \frac{m}{M} RT$

$$\rho_a = \frac{m}{V} = \frac{pRT}{M} \frac{pM}{RT}$$

$$= \frac{984 \times 10^2 \text{ Pa} \times 28.96 \times 10^{-3} \text{ kg mol}^{-1}}{8.314 \text{ J K}^{-1} \text{ mol}^{-1} \times 288 \text{ K}}$$

$$= \underline{\underline{1.19 \text{ kg m}^{-3}}}$$

- ⑤ The mass concentration of sea salt aerosol in marine air decreases exponentially with altitude with a scale height of 0.5 km. The average mass concentration of sea salt almost in surface air over the ocean is $10 \mu\text{g m}^{-3}$. The Earth is 70% ocean and the sea salt aerosol concentration over land is negligible. Calculate the total mass (in kg) of sea salt in the atmosphere if the mass of the sea salt is given as $m = Ap(0)h$

ans) $A = 4\pi R^2$, $R = 6.4 \times 10^6 \text{ m}$

$$\rho(0) = \rho_a = 1.19 \text{ kg m}^{-3} \times 10 \times 10^{-6} \times 10^{-3} \text{ kg m}^{-3}$$

$$= 10^{-8} \text{ kg m}^{-3}$$

$$h = 0.5 \text{ km} = 500 \text{ m}$$

$$m = A \rho(0) h$$

$$= (7) \pi (6.4 \times 10^6 \text{ m})^2 \times 10^{-8} \text{ kg m}^{-3} \times 500 \text{ m}$$

$$= 1.8 \times 10^9 \text{ kg}$$

$$= \underline{1.8 \times 10^9 \text{ kg}} \text{ of sea salt.}$$

③ Give a rough order of magnitude for the number of molecules present in a typical lightning bolt.

$$V = \frac{4}{3} \pi r^3 = \frac{4}{3} \pi (2.5 \times 10^3 \text{ m})^3 = 2.6 \times 10^{10} \text{ m}^3$$

Assuming the particle is water

$$m = \rho V = 10^3 \text{ kg m}^{-3} \times 2.6 \times 10^{10} \text{ m}^3 = 2.6 \times 10^{13} \text{ kg}$$

$$m = 2.6 \times 10^{13} \text{ kg}$$

$$m = 2.6 \times 10^{13} \text{ kg}$$

$$M = 18 \times 10^{-3} \text{ kg mol}^{-1}$$

$$n = \frac{m}{M} = \frac{2.6 \times 10^{13} \text{ kg}}{18 \times 10^{-3} \text{ kg mol}^{-1}} = 1.4 \times 10^{16} \text{ mol}$$

$$N = n \times 6.02 \times 10^{23} = 8.4 \times 10^{39}$$

$$N = 8.4 \times 10^{39}$$

no. of molecules = 8.4×10^{39}

$$N = 8.4 \times 10^{39}$$

roughly 10^{40} molecules are present.

Ch 1. Additional questions

- ① Oxygen has a fixed mixing ratio in the atmosphere. How would you expect its number density measured in surface air to vary between day and night? How would you expect its partial pressure measured in surface air to vary between day and night?

ans) During the day there is photosynthetic activity that produces O_2 . Hence its number density and its partial pressure is higher during the day and lower during the night.

- ② Give a rough order of magnitude for the number of molecules present in a typical $1 \mu m$ aerosol particle.

ans) $r = 0.5 \times 10^{-6} m$
 $V = \frac{4}{3} \pi r^3 = 0.524 \times 10^{-18} m^3$

Assuming the particle is water

$$m = \rho V = 10^3 kg m^{-3} \times 0.524 \times 10^{-18} m^3$$

$$= 0.524 \times 10^{-15} kg$$

$$= 5.24 \times 10^{-16} kg$$

$$M = 18 \times 10^{-3} kg mol^{-1}$$

$$n = \frac{m}{M} = \frac{5.24 \times 10^{-16} kg}{18 \times 10^{-3} kg mol^{-1}}$$

$$= 2.91 \times 10^{-14} moles$$

$$no. of molecules = 2.91 \times 10^{-14} \times 6.022 \times 10^{23}$$

$$= 1.75 \times 10^{10} molecules.$$

roughly 10^{10} molecules are present.

14/09 Mid Sem ?

1) Q.5. What is the scale height of Earth's atmosphere taking $T = 250\text{K}$ & $g = 10\text{ms}^{-2}$?

ans) $H = \frac{RT}{Mg}$ $M_a = 29 \times 10^{-3}\text{kg mol}^{-1}$

$$= \frac{8.314 \times 250\text{K}}{29 \times 10^{-3}\text{kg mol}^{-1} \times 10\text{ms}^{-2}}$$

$$= 7.2\text{ km}$$

20/09
 5) Q.6.

$$n_{\text{neon}} = C_{\text{neon}} n_{\text{air}}$$

$$n_{\text{air}} = \frac{p_{\text{air}}}{RT} \quad n_{\text{air}} \propto \frac{1}{T} \Rightarrow n_{\text{neon}} \propto \frac{1}{T}$$

At night $T \downarrow \therefore n_{\text{neon}} \uparrow$

$$p_{\text{neon}} = C_{\text{neon}} p_{\text{air}}$$

$$p_{\text{air}} = \frac{\text{weight of an overhead column}}{\text{area}}$$

$\downarrow$
 assumed constant

$\therefore p_{\text{neon}}$ doesn't change.

doesn't change
 between day & night

20

6) Q.7.

- (i) intercomparison of abundance
- (ii) rate of rxn ; optical scattering or absorption
- (iii) exposure relationship.
- (iv) phase transformation (cloud formation)

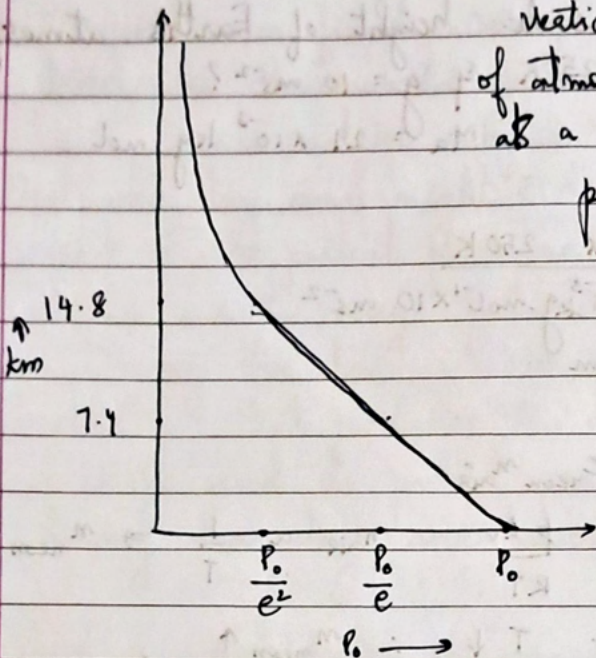
6) Q.8.

$$n_x = C_x n_a$$

$$\frac{n_x M_x}{A_v} = \frac{C_x p_{\text{air}} M_x}{RT A_v} \Rightarrow p_x = \frac{C_x p_{\text{air}} M_x}{RT}$$

c) Q.11.

vertical profile
 of atmospheric pressure
 at a fraction of surface
 pressure (P_0)



EES202

T4

11/10

- ① Buoyant motions in the atmosphere associated with local differences in heating can be vigorous. Consider a black parking lot where surface air temperature ($T=303K$) is slightly warmer than that of the surrounding area ($T'=300K$). What is the buoyant acceleration of the air over the parking lot?

ans) buoyant acceleration, $\alpha_b = \frac{\rho' - \rho}{\rho} g = \frac{T - T'}{T'} g$

$$\alpha_b = \frac{303 - 300}{300} (g) = \underline{0.1 \text{ m s}^{-2}}$$

- ② When an object set to target from a distance of ¹⁰⁰⁰ km at 2000 km h^{-1} , what is the displacement can be caused to the object due to Coriolis force? (latitude 42°)

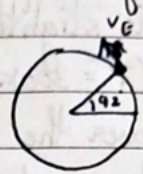
ans) Displacement $\Delta y = \frac{w(\Delta x)^2 \sin \lambda}{V}$ $\Delta x = 1000 \text{ km}$
 $V = 2000 \text{ km h}^{-1}$
 $V = \omega R \text{ or } \omega = V/R$
 $\lambda = 42^\circ$
 $A = 42^\circ$
 $= \frac{V(\Delta x)^2 \sin \lambda}{R \omega}$
 $= \frac{(\Delta x)^2 \sin \lambda}{R}$
 $= \frac{(1000 \text{ km})^2 \sin 42^\circ}{6400 \text{ km}}$
 $= \frac{10^6 \times \sin 42^\circ}{6400} \text{ km}$
 $= \frac{10^6 \times 0.67}{6400} \text{ km}$
 $= \underline{1.05 \times 10^2 \text{ km}}$

(3) Consider an observer fixed in space and watching the Earth rotate. What is the translational velocity of an object on the Earth at latitude 42° from the perspective of observer?

ans)

translation velocity,

$$v_E = \frac{2\pi R \cos \theta}{t}$$



fixed observer

$$= \frac{2 \times 3.14 \times 6400 \text{ km} \times \cos 42^\circ}{24 \text{ hr}}$$

$$= \frac{2 \times 3.14 \times 6400 \times 10^3 \text{ m} \times \cos 42^\circ}{24 \times 60 \times 60 \text{ s}}$$

$$= \underline{\underline{3.44 \times 10^2 \text{ m s}^{-1}}}$$

(4) An air parcel with a density of 1.1 kg m^{-3} is surrounded by air with a density of 1.2 kg m^{-3} . If the volume of air parcel is 5 m^3 , calculate the buoyant force acting on the air parcel.

ans)

buoyant acceleration, $\cancel{a_b} = \frac{\rho' - \rho}{\rho} g$

Given $\rho = 1.1 \text{ kg m}^{-3}$ $\rho' = 1.2 \text{ kg m}^{-3}$
 $g = 10 \text{ m s}^{-2}$

Buoyant force, $F_b = m g = \rho' V g$
 $= (1.2 \text{ kg m}^{-3}) (5 \text{ m}^3) (10 \text{ m s}^{-2})$
 $= \underline{\underline{60 \text{ N}}}$

$$= 60 \text{ kg m s}^{-2}$$

$$= \underline{\underline{60 \text{ N}}}$$

- ⑤ An air parcel at sea level has a temperature of 300K . It rises to a height of 1.5 km under dry adiabatic condition. If the dry adiabatic lapse rate is 10 K km^{-1} , what will be the temperature of the air parcel at 1.5 km ?

ans) $T = 300\text{K}$ $T_z = ?$ $\Gamma = 10\text{ K km}^{-1}$

$z = 1.5\text{ km}$ $\tau_z = 15\text{ K}$

$$T_z = T - \tau_z = 300 - 15$$

$$= \underline{\underline{285\text{K}}}$$